

Classifying Regular and Irregular Polygons

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Polygons are classified according to their **number of sides**.

Names of Polygons

A **triangle** has 3 sides.

A **quadrilateral** has 4 sides.

A **pentagon** has 5 sides.

A **hexagon** has 6 sides.

Names of Polygons

A **heptagon** has 7 sides.

An **octagon** has 8 sides.

A **nonagon** has 9 sides.

A **decagon** has 10 sides.

Regular Polygon

A polygon is **regular** if all sides and all angles are congruent.

Irregular Polygon

A polygon is **irregular** if its sides or angles are not all congruent.

Examples

Example 1

A polygon has 3 equal sides and 3 equal angles.
Since all sides and angles are equal, the polygon
is **regular**.

Example 2

A polygon has 5 sides of different lengths.
Since the sides are not all equal, the polygon
is **irregular**.

Example 3

A polygon has 5 equal sides, but one angle is different from the others.

Since all angles are not equal, the polygon is **irregular**.

Practice Exercises

Practice 1

A polygon has 4 sides where all sides are equal and all angles are equal.

Classify the polygon as **regular** or **irregular**.

Practice 2

A polygon has 6 equal sides and 6 equal angles.
Classify the polygon as **regular** or **irregular**.

Practice 3

A polygon has 7 sides, where two sides are longer than the others.

Classify the polygon as **regular** or **irregular**.

Practice 4

A polygon has 8 equal sides and equal angles.
Classify the polygon as **regular** or **irregular**.

Practice 5

A polygon has 9 sides of varying lengths.
Classify the polygon as **regular** or **irregular**.

Practice 6

A polygon has 10 equal sides and equal angles.
Classify the polygon as **regular** or **irregular**.

Practice 7

A polygon has 4 sides where the side lengths are not all equal.

Classify the polygon as **regular** or **irregular**.

Activities

Activity 1

A polygon has 3 sides where one side is longer than the other two.

Classify the polygon as **regular** or **irregular**.

Activity 2

A polygon has 4 sides and four equal angles, but the sides are not all equal.

Classify the polygon as **regular** or **irregular**.

Activity 3

A polygon has 5 equal sides and 5 equal angles.
Classify the polygon as **regular** or **irregular**.

Activity 4

A polygon has 6 sides of different lengths.
Classify the polygon as **regular** or **irregular**.

Activity 5

A polygon has 7 equal sides and 7 equal angles.
Classify the polygon as **regular** or **irregular**.

Activity 6

A polygon has 8 sides where several sides have different lengths.

Classify the polygon as **regular** or **irregular**.

Activity 7

A polygon has 9 equal sides and equal angles.
Classify the polygon as **regular** or **irregular**.

Activity 8

A polygon has 10 sides of varying lengths.
Classify the polygon as **regular** or **irregular**.

Activity 9

A polygon has 4 equal sides but one angle differs from the others.

Classify the polygon as **regular** or **irregular**.

Activity 10

A polygon has 5 sides where all side lengths are different.

Classify the polygon as **regular** or **irregular**.

Thank you!